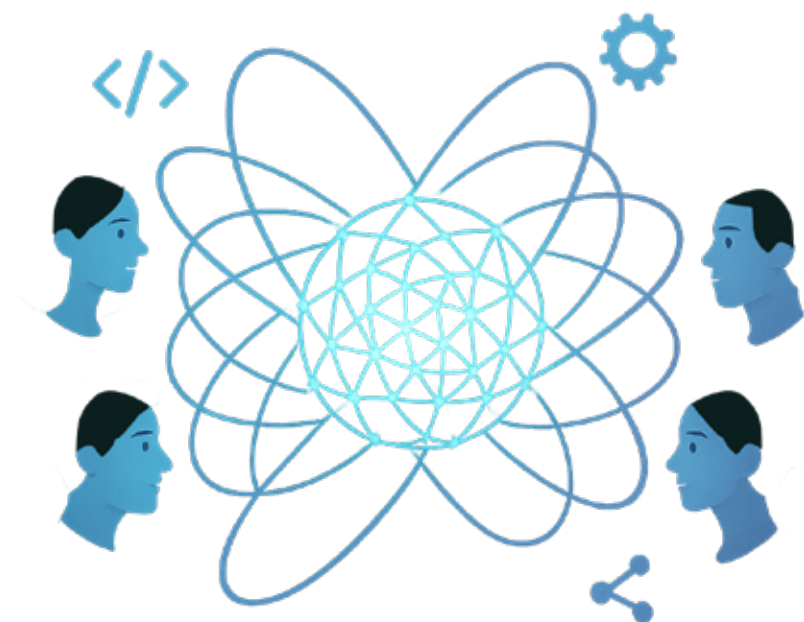


How Open Source is Augmenting Quantum Machine Learning

Quantum machine learning is a subject of huge interest today as it promises to solve the more complex problems we face. Open source is helping it cross global boundaries so that teams across the world can collaborate to improve lives. What's more, India has the potential to be a leader in this field.



Quantum computing and machine learning are transforming the way we work, learn and solve problems. Together they create quantum machine learning (QML), a rapidly growing technology that allows advanced data processing and improved decision making. Unlike classical machine learning, which powers recommendation engines, fraud detection systems, voice assistants and so many other products and services we see around us, quantum machine learning can tackle problems that are far too complex even for the fastest conventional computers.

Quantum computers use qubits (unlike binary bits), which can be 0, 1, or both simultaneously. They can

process multiple possibilities at once, which helps them solve problems much faster than classical computers that use sequential methods. For instance, think about how to arrange the optimal routes for hundreds of delivery trucks in a city. A traditional computer would evaluate each route one at a time. Even with the best processors and resources, this takes a significant amount of time. A quantum computer can analyse numerous paths at once and get answers much more quickly.

Open source platforms such as Qiskit, Cirq and PennyLane provide users with accessible quantum machine learning capabilities. A student in Bengaluru can run quantum neural network simulations on a simulator,

researchers in Zurich can use cloud-based quantum hardware to validate results and Toronto-based startups can work on financial models. This open participation facilitates worldwide collaboration between researchers engaging in joint experimental work and innovative projects.

Quantum machine learning is no longer only a research curiosity. It has progressed into a practical discipline that attracts worldwide participation.

The rise of quantum machine learning

Data and pattern recognition are the building blocks of machine learning. The more data a model gets, the better it becomes. But as data grows, so do the challenges. Classical computers struggle when it comes to very complex tasks like large scale optimisation or simulating molecules at the molecular level. There are limits to even the fastest supercomputers. This is where quantum computing comes into play.

Quantum computing emerged as a concept during the 1980s. Richard Feynman and David Deutsch, among other prominent physicists, demonstrated that quantum mechanics could help computers perform calculations that classical systems would never be able to achieve. Theoretical development of quantum computing remained limited because scientists struggled to build stable quantum systems. The development of qubits, error correction methods and quantum gates in recent years has